Chapter 116

Resolve Connect

This chapter details the single node found in the Resolve Connect category, available only in standalone Fusion Studio.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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External Matte Saver [EMS]

The External Matte Saver node

NOTE: The Resolve Connect category and External Matte Saver node are available only in Fusion Studio.

External Matte Saver Node Introduction

The External Matte Saver node renders multiple mattes into multiple channels of an EXR file. This file is intended to import into DaVinci Resolve's Color page as an efficient way to deliver multiple mattes for color grading. To perform the same operation with a traditional Saver node, a Channel Boolean is needed to place each matte into a channel, and then name that channel in the Saver. It requires a bit more of a set up. This node streamlines the process by providing multiple inputs and naming the channel based on the node.

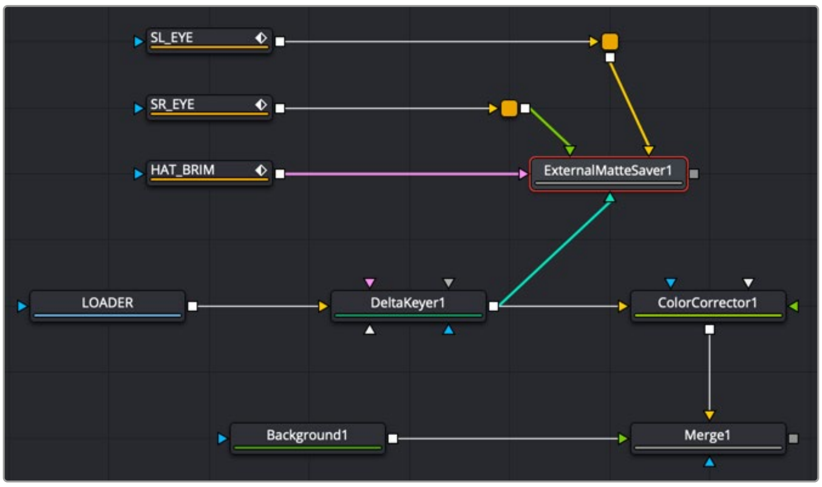
Inputs

By default, the node provides a single input for a 2D image you want to save as a matte.

Input: Although initially there is only a single orange input for a matte to connect, the Inspector provides an Add button for adding additional inputs. Each input uses a new color, but all accept 2D RGBA images.

Basic Node Setup

The External Matte Saver node can be placed at the end of any node tree branch for saving mattes. Below, it is connected to the Delta Keyer as well as three other mattes. Each matte beyond the initial one is connected by clicking the Add button in the Inspector first, and then a new input is provided on the node.



An External Matte Saver node added as a separate branch in a node tree to render the mattes

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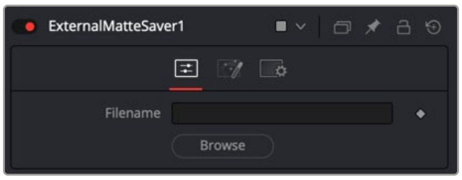
Inspector



The External Matte Saver Controls tab

Controls Tab

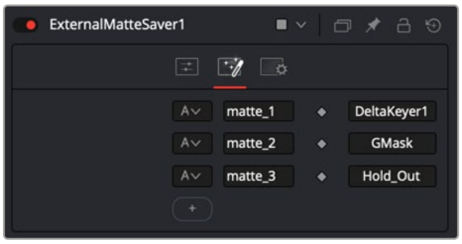
The Controls tab is used to name the saved file and determine where on your hard drive the file is stored.

Filename

Enter the name you want to use for the EXR file in the Filename field. At the end of the name, append the .exr extension to ensure that the file is saved as an EXR file.

Browse

Clicking the Browse button opens a standard file browser window where you can select the location to save the file.



The External Matte Saver Mattes tab

Mattes Tab

The Mattes tab is where you set up the number of mattes saved in the file, the name for each channel, and the RGBA channels saved from each input.

Channels menu

The Channels menu allows you to select which channels are saved in the matte. You can choose the alpha channel, the RGB channels, or the RGBA channels.

Channels Name

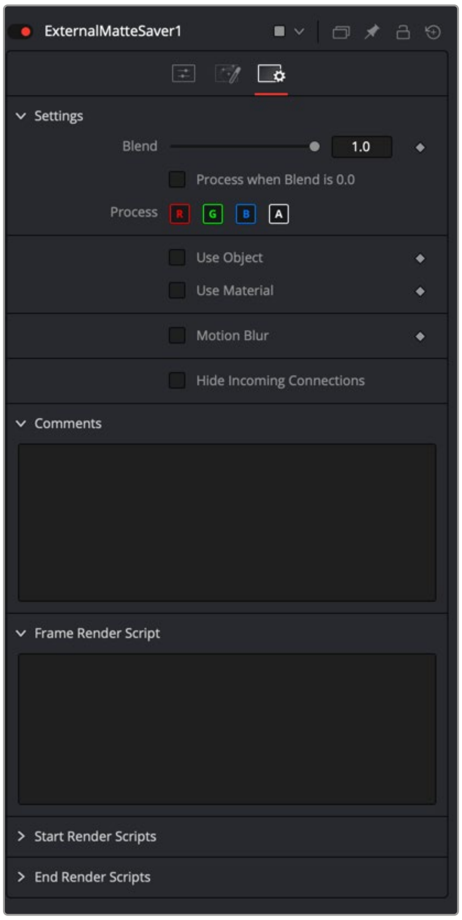
The Channels Name field allows you to customize the name of the matte channel you are saving. This name is displayed in DaVinci Resolve's Color page.

Node Name

The Node Name field displays the source of the matte. This is automatically populated when you connect a node to the input.

Add

Clicking the Add button adds an input on the node and another set of fields for you to configure and name the new matte channel.



The External Matte Saver Settings tab

Settings Tab

The Settings Tab in the Inspector is similar to settings found in the Saver tool. The controls are consistent and work the same way as the Settings in other tools.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

> For more information on the Coverage and Background Color channels, *see Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.